

Revision — fractions and their division

The four operations on fractions gathered before the control work.

LESSON 34–35

2 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 7 CONSOLIDATION

LESSON CLOCK

20'

20'

25'

12'

3

The four operations side by side

20 min

Mixed numbers

20 min

Word problems

25 min

The errors that remain

12 min

Homework

3 min

LEARNING OBJECTIVES

- Choose the right method for each of the four operations.
- Work fluently with mixed numbers.
- Solve a word problem in three lines.
- Identify and repair the errors that remain.

TEXTBOOK

National

pp. 94–98

Cambridge

Stage 7, pp. 66–80

Workbook

Revision 7

01

Explanation

The four operations side by side

OPERATION	WHAT IT NEEDS	EXAMPLE
addition	a common denominator	$\frac{2}{3} + \frac{1}{4} = \frac{11}{12}$
subtraction	a common denominator	$\frac{5}{6} - \frac{1}{3} = \frac{1}{2}$
multiplication	nothing — multiply across	$\frac{2}{3} \cdot \frac{3}{5} = \frac{2}{5}$
division	the reciprocal of the divisor	$\frac{2}{3} \div \frac{4}{5} = \frac{5}{6}$

ONLY + AND − NEED A COMMON DENOMINATOR

Finding one before multiplying or dividing is not wrong, only wasted work — and it is the single commonest habit that slows this topic down.

Mixed numbers

OPERATION	BEST METHOD	EXAMPLE
adding	wholes and fractions separately, or convert	$1\frac{1}{2} + 2\frac{1}{3} = 3\frac{5}{6}$
subtracting	convert if borrowing is needed	$3\frac{1}{4} - 1\frac{1}{2} = 1\frac{3}{4}$
multiplying	always convert first	$1\frac{1}{2} \cdot 2 = 3$
dividing	always convert first	$2\frac{1}{4} \div 1\frac{1}{2} = 1\frac{1}{2}$

NEVER MULTIPLY THE WHOLE PARTS AND THE FRACTIONS SEPARATELY

$1\frac{1}{2} \cdot 2\frac{1}{2}$ is not $2\frac{1}{4}$. Convert to $\frac{3}{2} \cdot \frac{5}{2}$, which is $\frac{15}{4} = 3\frac{3}{4}$.

Word problems

PROBLEM	OPERATION	ANSWER
$\frac{3}{4}$ of 24 pupils are present	multiply	18
9 m of rope in $\frac{3}{4}$ m pieces	divide	12
$\frac{2}{3}$ of a tank is 20 l	divide	30 l
$\frac{1}{2}$ l shared between 4	divide	$\frac{1}{8}$ l
$\frac{1}{4}$ added to $\frac{1}{3}$ of a day	add	$\frac{7}{12}$

THREE LINES EVERY TIME

Name the operation, do the arithmetic, answer in words with units. The middle line is the easiest of the three and carries the fewest marks.

The errors that remain

ERROR	LOOKS LIKE	CORRECT
adding denominators	$\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$	$\frac{5}{6}$
common denominator before ×	extra work	multiply across
turning over the wrong fraction	$\frac{2}{3} \div \frac{4}{5} = \frac{3}{2} \cdot \frac{4}{5}$	$\frac{2}{3} \cdot \frac{5}{4}$
mixed number multiplied in parts	$1\frac{1}{2} \cdot 2\frac{1}{2} = 2\frac{1}{4}$	$3\frac{3}{4}$
answer not simplified	$\frac{12}{18}$	$\frac{2}{3}$

02 Worked examples

EXAMPLE 1 Compute $1\frac{1}{2} \cdot 2\frac{1}{2}$.

1

Convert both: $\frac{3}{2}$ and $\frac{5}{2}$.

2

$\frac{3}{2} \cdot \frac{5}{2} = \frac{15}{4}$

3

$= 3\frac{3}{4}$

ANSWER

$3\frac{3}{4}$

EXAMPLE 2 Compute $3\frac{1}{4} - 1\frac{1}{2}$.

① Convert: $\frac{13}{4} - \frac{3}{2}$.

② Common denominator 4: $\frac{13}{4} - \frac{6}{4}$.

③ $= \frac{7}{4} = 1\frac{3}{4}$

ANSWER

$$1\frac{3}{4}$$

EXAMPLE 3 $\frac{2}{3}$ of a tank is 20 litres. Find its capacity.

① A part is given, the whole is wanted: divide.

② $20 \div \frac{2}{3} = 20 \cdot \frac{3}{2}$

③ $= 30$ litres.

ANSWER

30 litres

03 Interactive model

Put one addition, one multiplication and one division on the board unlabelled and ask which needs a common denominator; the answer separates fluency from habit.

Which method does this one need?

$\frac{2}{3} + \frac{1}{4}$ needs:

a common denominator

the reciprocal

nothing

converting

$\frac{2}{3} \cdot \frac{3}{5}$ needs:

a common denominator

the reciprocal

multiplying across

converting

$\frac{2}{3} \div \frac{4}{5}$ needs:

a common denominator

the reciprocal of the divisor

nothing

squaring

$\frac{1}{2} + \frac{1}{3}$ equals:

$$\frac{2}{5}$$

$$\frac{5}{6}$$

$$\frac{1}{6}$$

$$\frac{2}{6}$$

$1\frac{1}{2} \cdot 2\frac{1}{2}$ equals:

$$2\frac{1}{4}$$

$$3\frac{3}{4}$$

$$3$$

$$2\frac{1}{2}$$

$3\frac{1}{4} - 1\frac{1}{2}$ equals:

$$1\frac{3}{4}$$

$$2\frac{1}{4}$$

$$1\frac{1}{4}$$

$$2\frac{3}{4}$$

2 3 of a tank is 20 l; the tank holds:

$$13\frac{1}{3} \text{ l}$$

30 l

40 l

60 l

12 18 as an answer is:

finished

not simplified

wrong

improper

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Revision	Takrorlash	Повторение
Common denominator	Umumiy maxraj	Общий знаменатель
Reciprocal	Teskari son	Обратное число
Mixed number	Aralash son	Смешанное число
To cancel	Qisqartirish	Сокращать
Lowest terms	Qisqarmas ko'rinish	Несократимый вид
Word problem	Matnli masala	Текстовая задача
Check	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Which operations need a common denominator?

all four

+ and –

× and ÷

none

To divide by a fraction:

find a common denominator

multiply by the reciprocal

multiply across

subtract

Before multiplying mixed numbers you should:

multiply the wholes

convert to improper fractions

add them

round them

$\frac{1}{2} + \frac{1}{3}$ equals:

$$\frac{2}{5}$$

$$\frac{5}{6}$$

$$\frac{1}{5}$$

$$\frac{2}{6}$$

9 m of rope in $\frac{3}{4}$ m pieces gives:

6.75

12

9

3

An answer of $\frac{12}{18}$ should be written:

$$\frac{12}{18}$$

$$\frac{2}{3}$$

$$\frac{6}{9}$$

0.67

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up**7 PROBLEMS**

1. $\frac{2}{3} + \frac{1}{4}$

ANSWER $\frac{11}{12}$

2. $\frac{5}{6} - \frac{1}{3}$

ANSWER $\frac{1}{2}$

3. $\frac{2}{3} \cdot \frac{3}{5}$

ANSWER $\frac{2}{5}$

4. $\frac{2}{3} \div \frac{4}{5}$

ANSWER $\frac{5}{6}$

5. $\frac{1}{2} + \frac{1}{3}$

ANSWER $\frac{5}{6}$

6. $\frac{3}{4}$ of 24

ANSWER 18

7. 9 m in $\frac{3}{4}$ m pieces

ANSWER 12

Medium · the standard question

7 PROBLEMS

1. $1\frac{1}{2} + 2\frac{1}{3}$

ANSWER $3\frac{5}{6}$

2. $3\frac{1}{4} - 1\frac{1}{2}$

ANSWER $1\frac{3}{4}$

3. $1\frac{1}{2} \cdot 2\frac{1}{2}$

ANSWER $3\frac{3}{4}$

4. $2\frac{1}{4} \div 1\frac{1}{2}$

ANSWER $1\frac{1}{2}$

5. $\frac{2}{3}$ of a tank is 20 l: the tank

ANSWER 30 l

6. $\frac{1}{2}$ l shared between 4

ANSWER $\frac{1}{8}$ l

7. $\frac{1}{4} + \frac{1}{3}$

ANSWER $\frac{7}{12}$

Hard · stretch and reason

7 PROBLEMS

1. $2\frac{1}{3} \cdot 1\frac{2}{7}$

ANSWER 3

2. $4\frac{1}{2} \div \frac{3}{4}$

ANSWER 6

3. $\frac{3}{4} + \frac{2}{3} - \frac{1}{2}$

ANSWER $\frac{11}{12}$

4. $\frac{5}{8} \div \frac{5}{6} \cdot \frac{2}{3}$

ANSWER $\frac{1}{2}$

5. A recipe needs $\frac{3}{4}$ kg for 6 people: for 8

ANSWER 1 kg

6. Which is larger, $\frac{2}{3} \div \frac{3}{4}$ or $\frac{2}{3} \cdot \frac{3}{4}$?

ANSWER The division

7. Why is a common denominator not needed for $\times$?

ANSWER The parts are recut by the multiplication itself

07 Homework

Homework — 5 tasks

Name the operation before starting, and simplify every answer.

1. Compute $\frac{3}{5} + \frac{1}{4}$ and $\frac{7}{8} - \frac{1}{2}$.

2. Compute $\frac{4}{9} \cdot \frac{3}{8}$ and $\frac{4}{9} \div \frac{8}{3}$.
3. Compute $2\frac{1}{2} \cdot 1\frac{1}{5}$.
4. Compute $3\frac{3}{4} \div 1\frac{1}{4}$.
5. $\frac{5}{8}$ of a barrel is 25 litres. Find its capacity.